

HR Attrition Analytics Dashboard

From messy workforce records to an executive decision tool

THE CHALLENGE

NovaTech needed a clear way to identify where employee departures were concentrated, which workforce conditions were associated with elevated attrition, and which HR actions deserved priority.

DATASET	TOOLS	OUTCOME
1,468	Excel	15.4%
clean employee records	PivotTables + formulas	company attrition rate

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End-to-end Excel analytics portfolio project | Version 1.0 | July 2026

PROJECT AT A GLANCE

1. Executive Overview

This case study documents how a raw HR dataset was transformed into a structured, validated, and decision-focused Excel dashboard. The project demonstrates the complete analytics lifecycle: business framing, data cleaning, feature engineering, KPI design, PivotTable analysis, dashboard development, insight generation, and actionable recommendations.

HEADCOUNT	DEPARTURES	ACTIVE	ATTRITION
1,468	226	1,242	15.4%
employees analyzed	recorded exits	employees retained	company benchmark

Business outcome

DECISION VALUE

The analysis identified overtime, first-year tenure, low job satisfaction, and selected sales-related roles as the clearest retention priorities. It also separated high-rate small segments from high-volume drivers, reducing the risk of overreacting to unstable percentages.

Skills demonstrated

Capability	Evidence in the project
Business analysis	Translated a broad turnover concern into measurable questions and acceptance criteria.
Data preparation	Cleaned inconsistent records, handled missing categories, validated identifiers, and preserved raw data lineage.
Excel analytics	Used structured tables, helper columns, PivotTables, calculated rates, charts, slicers, and KPI cards.
Data storytelling	Designed an executive dashboard and converted findings into prioritized HR actions.
Governance	Documented definitions, limitations, assumptions, quality checks, and reproducibility steps.

BUSINESS FRAMING

2. Business Problem and Objectives

The problem

NovaTech had employee records but no integrated analytical view of attrition. Static reporting showed how many employees had left, but it did not clearly explain where risk was concentrated, which segments drove departure volume, or which issues HR could address first.

Primary objective

OBJECTIVE

Build an interactive Microsoft Excel dashboard that enables HR leaders to monitor attrition, compare workforce segments, identify practical retention risks, and support evidence-based intervention planning.

Questions the dashboard needed to answer

Theme	Core question
Overall exposure	What is the company attrition rate and how many employees have left?
Organization	Which departments and roles combine elevated rates with meaningful departure volume?
Work conditions	How strongly is overtime associated with attrition?
Employee lifecycle	Are younger and early-tenure employees leaving more often?
Experience	How do satisfaction and work-life indicators relate to retention?
Compensation	Do lower income groups show higher attrition, and are sample sizes sufficient?
Action	Which interventions should HR prioritize over the next 90 days?

Scope discipline

The project is descriptive and diagnostic, not predictive. It identifies patterns and associations within a static workforce snapshot. It does not claim causation, estimate future exits, or connect to a live HR information system.

DATA FOUNDATION

3. Dataset, Cleaning, and Feature Engineering

RAW RECORDS	CLEAN RECORDS	FIELDS	REMOVED
1,488	1,468	37	20
before cleaning	analysis-ready	including derived columns	invalid/duplicate records

Workbook architecture

Worksheet	Role
Raw_Data	Untouched source records retained for lineage and auditability.
Cleaned_Data	Validated analysis table with standardized categories and helper fields.
Data_Dictionary	Definitions, data types, code meanings, and derived-field logic.
Pivot_Tables	Reusable calculation layer supporting dashboard charts and KPIs.
Dashboard	Executive-facing interactive report.
Insights	Evidence-based findings written in business language.
Recommendations	Prioritized HR actions linked to the strongest signals.

Key preparation steps

Step	What was done	Why it mattered
Duplicate and ID checks	Validated EmployeeNumber uniqueness and removed invalid duplicate rows.	Prevents overstated headcount and attrition.
Missing-category handling	Preserved Unknown values for department and gender rather than silently deleting them.	Keeps totals reconcilable and exposes source-quality gaps.
Type validation	Confirmed numeric, categorical, and ordinal fields.	Protects PivotTable calculations and sorting.
Constant-field removal	Excluded non-analytical fields from the cleaned analytical table.	Reduces clutter without altering source data.
Derived features	Created AttritionFlag, AgeGroup, IncomeGroup, TenureGroup, and SatisfactionGroup.	Makes rate calculations and segment comparisons repeatable.

Analytical design choice

RATE BEFORE COUNT

Department and role comparisons use attrition rate as well as departure count. Rates normalize for different segment sizes, while counts reveal operational volume. Both are required for responsible prioritization.

TECHNICAL WORKFLOW

4. Analytical Method

The project followed a controlled sequence so that every dashboard result could be traced back to the cleaned data table.

Stage	Method	Output
1. Define	Translate stakeholder needs into measurable KPIs and business questions.	BRD and analysis plan
2. Inspect	Review structure, blanks, duplicates, category consistency, and data ranges.	Quality issue log
3. Clean	Standardize and validate records while preserving an untouched raw sheet.	Cleaned_Data
4. Engineer	Create interpretable grouping and binary helper fields.	Analysis-ready features
5. Analyze	Build PivotTables using consistent denominators and segment logic.	Validated calculation layer
6. Visualize	Create KPI cards, charts, labels, slicers, and executive hierarchy.	Interactive dashboard
7. Interpret	Compare rates, counts, sample sizes, and practical actionability.	Evidence-based insights
8. Recommend	Prioritize actions by impact, feasibility, ownership, and timing.	90-day HR action plan

Core KPI definitions

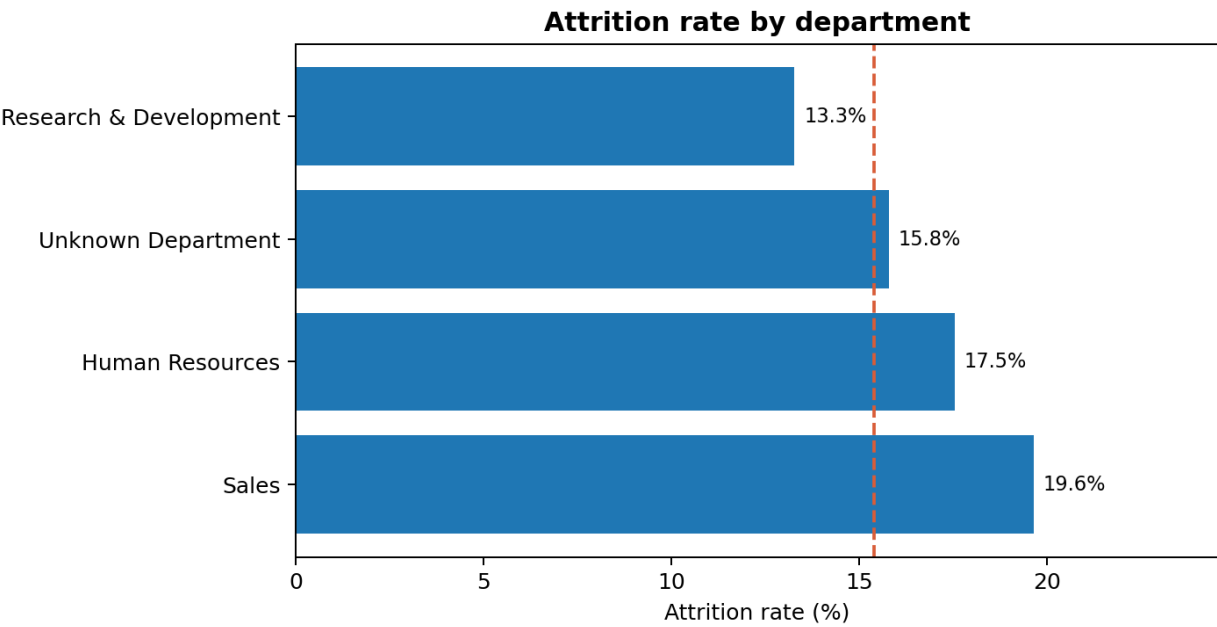
KPI	Definition	Result
Total headcount	Count of valid employee records	1,468
Departures	Sum of AttritionFlag	226
Active employees	Headcount minus departures	1,242
Attrition rate	Departures divided by headcount	15.4%
Average age	Mean employee age	36.4 years
Average monthly income	Mean MonthlyIncome	9,441
Average tenure	Mean YearsAtCompany	3.8 years

DASHBOARD OUTPUT

5. Executive Dashboard Design

The dashboard was designed for a 30-second executive scan followed by deeper filtering. The top row answers “How large is the issue?” The middle section explains “Where is it concentrated?” The lower views support “What should HR investigate?”

TOTAL	DEPARTURES	ACTIVE	AVG TENURE
1,468	226	1,242	3y 8m
workforce base	recorded exits	84.6% retained	years at company



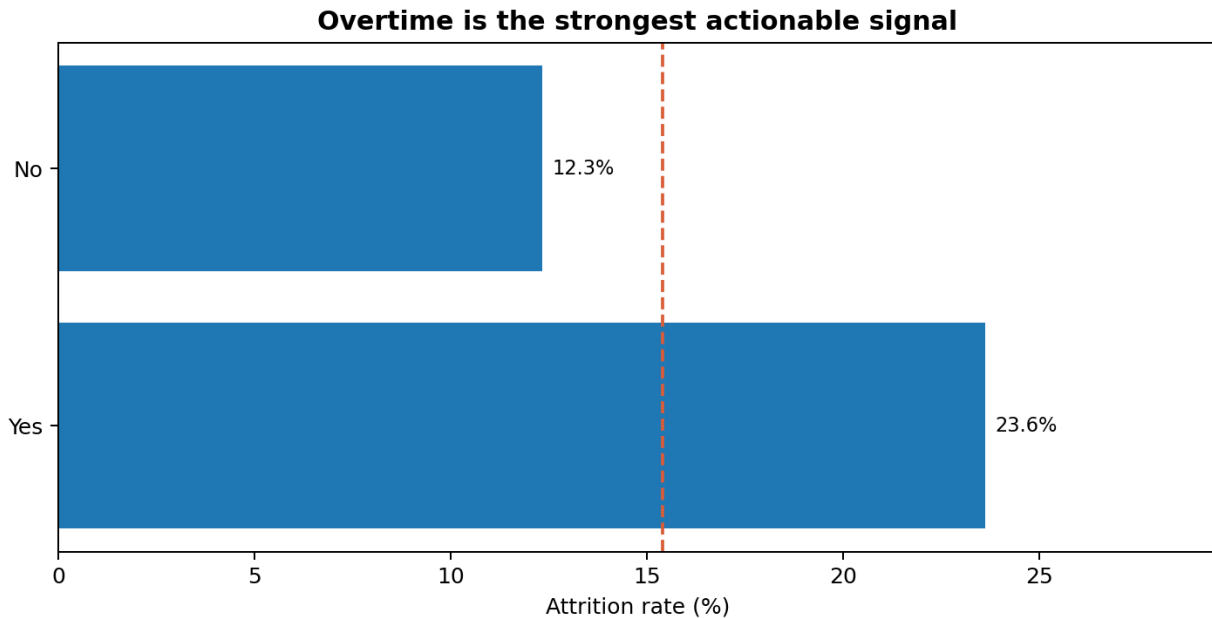
Design principles

Principle	Application
Decision-first hierarchy	KPIs and the highest-value risk signals appear before demographic detail.
Consistent metric logic	Charts emphasize attrition rate, with volume included in supporting tables.
Minimal visual noise	No 3D charts, unnecessary color variation, or decorative elements.
Action-oriented labeling	Titles state the business message rather than only naming the variable.
Interactive exploration	Slicers support controlled segment review without changing core definitions.

FINDINGS

6. Key Findings

Finding 1 - Overtime is the strongest actionable signal



Employees working overtime show a 23.6% attrition rate, compared with 12.3% for employees without overtime. The gap is large, the sample sizes are meaningful, and workload practices can be changed, making this the highest-priority management signal in the dataset.

Finding 2 - Sales combines high rate and high volume

Sales records the highest departmental attrition rate at 19.6% and contributes 87 departures. Human Resources is also above the company benchmark at 17.5%, but its smaller workforce means Sales represents the larger operational exposure.

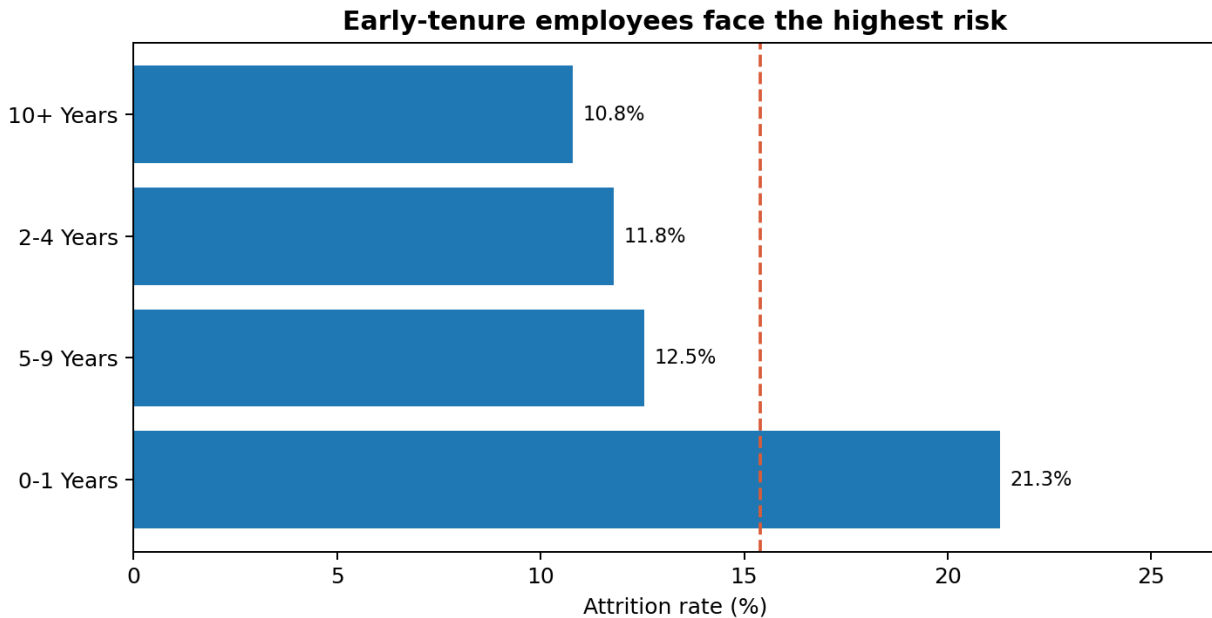
Finding 3 - Role-level risk requires sample-size judgment

Role	Employees	Departures	Rate	Interpretation
HR Business Partner	21	5	23.8%	Highest rate, but small base
Sales Representative	152	33	21.7%	High rate and meaningful volume
Research Scientist	154	30	19.5%	Elevated risk
Sales Executive	152	27	17.8%	Above benchmark
Research Director	161	13	8.1%	Lowest role rate

RISK PROFILE

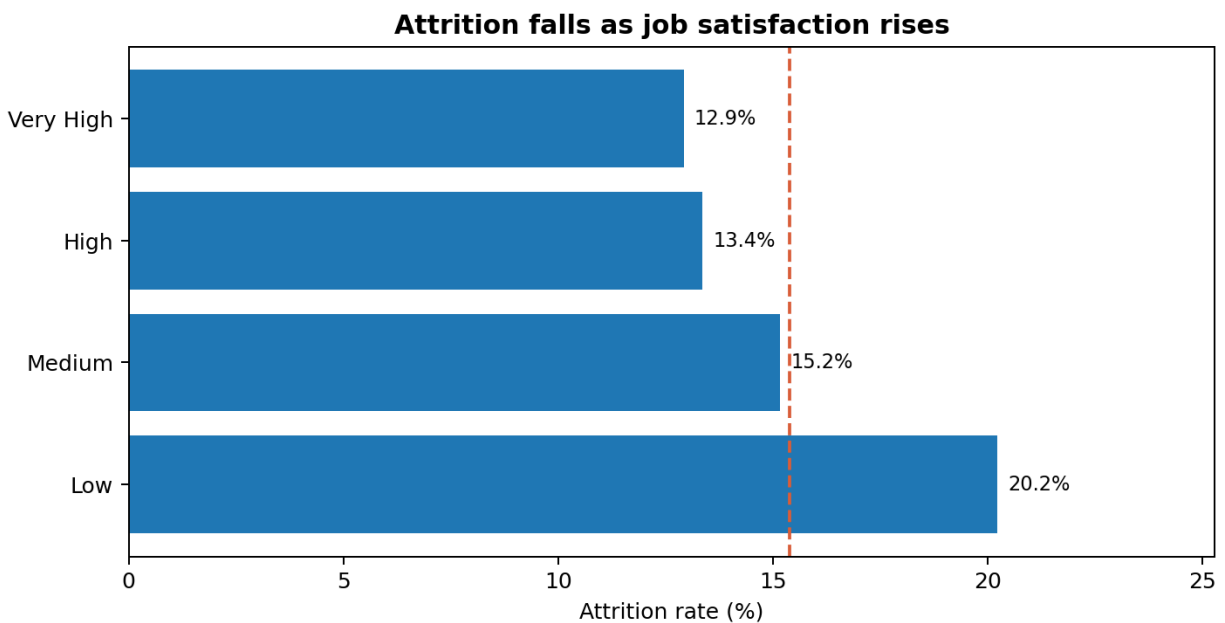
7. Employee Lifecycle and Experience Signals

Early tenure is the most vulnerable period



Employees with 0-1 years at the company have a 21.3% attrition rate. Risk declines sharply after the first year, supporting a targeted onboarding and early-career retention strategy rather than a broad intervention for all tenure groups.

Satisfaction shows a clear gradient



Low job satisfaction is associated with 20.2% attrition, compared with 12.9% among employees reporting very high satisfaction. The direction is consistent across the scale, making satisfaction useful as an early-warning and manager-conversation indicator.

BUSINESS ACTION

8. Recommendations and 90-Day Action Plan

Priority	Action	Owner	Timing	Success indicator
1	Review overtime concentration and redistribute workload in Sales and other high-exposure teams.	HR + Department Leaders	0-30 days	Reduced overtime exposure and attrition trend decline
2	Implement structured 30/60/90-day onboarding check-ins and first-year career conversations.	HR Operations	0-60 days	Lower 0-1 year attrition
3	Investigate working conditions in Sales Representative and HR Business Partner roles.	HR Business Partners	30-60 days	Documented root causes and action plans
4	Use job-satisfaction scores to trigger manager check-ins and development discussions.	People Managers	30-90 days	Fewer low-satisfaction employees and stronger retention
5	Establish monthly attrition governance and cohort reporting.	People Analytics	Monthly	Consistent trend monitoring and accountability

How priorities were chosen

Recommendations were ranked using four criteria: size of the rate gap, number of employees affected, actionability of the underlying issue, and stability of the segment evidence. This prevented the project from treating every high percentage as equally important.

IMPORTANT LIMITATION

The Low Income group records a high 29.4% attrition rate, but contains only 17 employees. It should be investigated, not presented as the primary company-wide driver without additional evidence.

Suggested governance thresholds

Indicator	Current baseline	Management trigger
Overall attrition rate	15.4%	Escalate sustained increase above baseline
Overtime attrition rate	23.6%	Prioritize if gap vs non-overtime remains >5 percentage points
First-year attrition rate	21.3%	Trigger onboarding review if above company average
Low job-satisfaction attrition	20.2%	Trigger manager intervention for persistent low scores
Unknown department records	19 employees	Resolve source mapping and prevent recurrence

QUALITY AND REFLECTION

9. Validation, Limitations, and Lessons Learned

Validation controls

Control	Evidence
Record reconciliation	Raw and cleaned record counts documented; removal count explained.
KPI reconciliation	Dashboard headcount, departures, active employees, and rate match PivotTable totals.
Definition consistency	Data dictionary defines each field, code, and derived group.
Rate denominator checks	Segment departures divided by total employees in the same segment.
Unknown-category transparency	Unknown department and gender values retained and separately reported.
Insight traceability	Each recommendation is linked to a measured finding and sample context.

Limitations

Limitation	Implication
Static snapshot	No monthly trend or seasonality analysis is possible.
Synthetic/portfolio dataset	Results demonstrate analytical technique rather than NovaTech operational truth.
Association, not causation	Overtime and satisfaction are correlated with attrition but are not proven causes.
No exit-reason data	Root causes require interviews, surveys, or exit records.
Small segments	Some high rates may be unstable and need cautious interpretation.
No cost model	The project does not quantify replacement cost or financial ROI of interventions.

Lessons learned

The most important analytical lesson was that the “highest rate” is not automatically the “highest priority.” Strong portfolio analysis must combine rate, volume, sample size, actionability, and business context. The project also reinforced that documentation and validation are part of the analytical product, not optional extras.

PORTFOLIO VALUE

10. Skills Demonstrated and Future Enhancements

Technical skills

Area	Skills demonstrated
Excel data preparation	Tables, filters, duplicate review, type checks, category standardization
Formulas and features	IF logic, grouping fields, binary flags, structured references
Analysis	PivotTables, attrition-rate calculations, segment comparisons, denominator control
Visualization	KPI cards, bar charts, labels, slicers, executive layout, conditional emphasis
Quality assurance	Reconciliation, source lineage, data dictionary, validation checklist
Communication	Executive summary, case study, recommendations, limitations, governance triggers

Future enhancements

Enhancement	Business value
Add monthly snapshots	Enable trend, cohort, seasonality, and intervention tracking.
Separate voluntary and involuntary exits	Improve interpretation and action ownership.
Add exit reasons and survey data	Move from association toward root-cause diagnosis.
Build a retention cost model	Estimate financial impact and prioritize interventions by ROI.
Migrate to Power BI	Support scalable refresh, richer drill-through, and broader distribution.
Develop a governed predictive model	Estimate individual risk only after data maturity, fairness review, and privacy controls.

Portfolio conclusion

PROJECT IMPACT

This project demonstrates more than chart creation. It shows how an analyst can structure an ambiguous business problem, protect data quality, build a transparent calculation layer, communicate risk responsibly, and translate findings into a realistic management plan.

Repository deliverables include the Excel workbook, dashboard image, Business Requirements Document, Data Dictionary, Project Methodology, Executive Report, Portfolio Case Study, validation checklist, and GitHub README.